

Do Market Models Fail When Confronted with Crisis?

Sep 30, 2026 · @ArmanBidarian · Code: [github.com/\[your-username\]/volatility-crisis-test](https://github.com/[your-username]/volatility-crisis-test)

Key Question

Generally, models we use to forecast the future are judged based on their ability to correctly predict. I trained a simple volatility model on data from 2005-2019, before testing its ability to predict volatility in Magnificent 7 stocks as well as the S&P 500 index from 2020-2025. I wanted to see how the accuracy of the model would compare against a naïve predictor (last month's volatility), especially when confronted with seemingly unpredictable events, such as the COVID-19 crash. To build this, I utilized Claude as a coding assistant and performed analysis on the results of a model using 21 years of daily prices for the selected securities.

Method

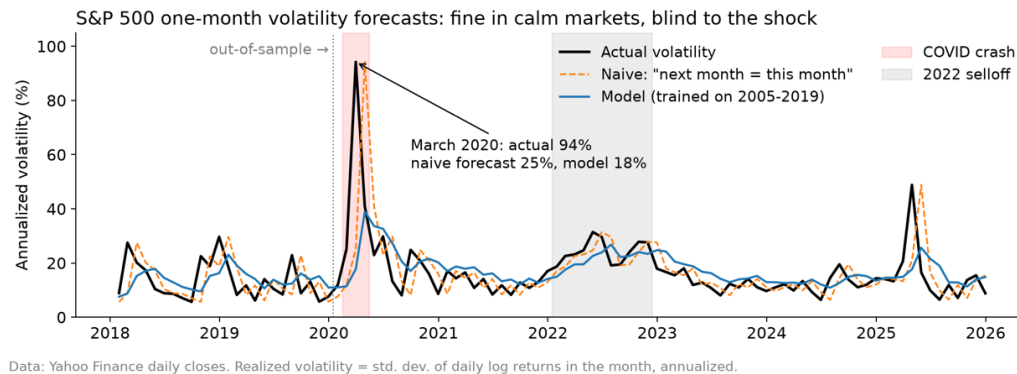
- **Target:** next month's realized volatility (annualized standard deviation of daily log returns within the month).
- **Model:** a linear regression (ordinary least-squares) based on six signals known at month-end: volatility (1M, 3M, 12M), returns (1M, 12M), and distance from 52-week high. Trained between 2005-2019 (1,248 stock-months, $R^2 = 0.59$).
- **Benchmark:** the naïve assumption: next month will look exactly the same as the previous one
- **Test:** both forecasts compared against empirics between 2020-2025, data not used to train the model, split into calm months, the COVID market crash (Feb-Apr 2020) and the 2022 selloff.

Results

Out-of-sample period	Naïve error	Model error	Model closer
Calm Months, 2020-2025	33%	29%	56% of months
COVID Crash, Feb-Apr 2020	59%	50%	38% of months
2022 selloff	26%	22%	50% of months

**Where error is the average absolute miss as a share of the actual volatility level, all 8 series pooled.*

During March 2020, S&P 500 volatility was at 94%, whilst the benchmark naïve index forecasts 25% and the model at 18%. March 2020 is 1.4% of all out-of-sample forecasts but 26% of the model's total squared error, twice its share of the naïve rule's error (13.5%).



Findings and Implications of the Test

1. *The model is more accurate at predicting observed volatilities in calm market settings.* In these times it beats the naïve benchmark, with no sign of overfitting.
2. *It is difficult for a complex of past price data to predict a sudden shock before it happens.* Both forecasts completely missed the COVID anomaly by 70 p.p. The same happened in April 2025 (tariff shock): actual volatility 49%, naïve forecast 21%, and model forecast 18%. These massive misses can largely be attributed to exogenous factors such as COVID, a Black Swan event which was largely outside what price history could capture.
3. *The model's strength is simultaneously its blind spot.* If we observe the regression coefficients used by the model, it is indicative of mean reversion: after an extreme month, it tends to pull the forecast back towards normal (indicated by the three volatility coefficients summing to 0.81, which is less than 1), qualitatively, this can be observed as the smoother blue line with less variation when compared with the benchmark and the observed outcome. This made it almost exactly right for April 2020 (model predicted 39% and 41% was observed). This mean reversion characteristic also indicates that the model is slow for predicting fundamental market shifts which establish a new standard volatility level. In 2022, considering the S&P 500, the naïve rule maintained an error of about 16%, the model kept pulling its forecast down toward the 2005-2019 average, meaning it was only closer in 4/12 months and had an error of 21%.
4. *Averages hide the tail.* A model can deliver fairly accurate results for an extended period but be broken by the one most critical month. An author such as Nassim Taleb would cite this as the inability of past events/data to predict the future, asserting that it is naïve to train these models on solely price data and expect it to somehow price in aggressive trade policy or a global pandemic.

Limitations

Seeing as the crises have already occurred, they were chosen with the quality of hindsight, and their samples are relatively small (only 3 months of data during the COVID crisis window). The Magnificent 7 used as a focus group omits silent evidence of the many companies which lost out during the crisis windows. They were obvious survivors and a continued study would likely include a fuller set of companies, varying market-cap, geographic location, and industry to train a more representative model. The model is never refit after its training period, it is therefore predicting 2025 with the weights established in 2019, to extend the study I would allow the model to be iteratively trained as it observes data within the testing period. This extends naturally into the next steps of the study. If I were to continue I would; refit the model each year, add option-implied volatility (VIX) as a more forward-looking signal, and expand the dataset to include a wider range of stocks.